MATH 463 - INTEGRAL EQUATIONS

LESSON 4

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OUTLINE

1. NUMERICAL QUADRATURE
   - Introduction
   - Trapezoidal Rule
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   - Exercises

Numerical Quadrature

For an integration, we seek to calculate the area under a given function say $f(x)$ within an interval $[a, b]$. The function may be a straight line, a curve etc. There are instances where calculating this area at a go becomes almost impossible to do, hence we rely on different approaches. One of them includes dividing the area into different segments, calculating the individual areas and then finally putting them together to make the full area(The resulting area is an approximation of the actual area). This is called **Numerical Integration** or **Quadrature** approach.

$$\int_a^b f(x)\ dx = V(f) + E(f)$$

$$\int_a^b f(x)\ dx = \sum_{i=0}^{n} \alpha_i f(x_i) + E(f)$$

Where $E(f)$ is the truncated error as a result of the approximation.

Let

$$\int_a^b f(x)\ dx \equiv \sum_{i=0}^{n} \alpha_i f(x_i) \triangle x$$

$$u(x) = f(x) + \lambda \int_a^b k(x,t) u(t)\ dt$$

$$u(x) = f(x) + \lambda \left[\sum_{j=0}^{n} \alpha_j k(x_i, t_j) u(t_j) \triangle t \right] \tag{1}$$

$$u(x) = f(x) + \lambda g(x) \left[\sum_{j=0}^{n} \alpha_j h(t_j) u(t_j) \triangle t \right] \tag{2}$$

Let look for the various $\{u_j\}_{j=0}^{n}$
Let $k(x_i, t_j)u(t_j) = k_{ij}u_j$
From (1)

$$u(x) = f(x) + \lambda\left[\sum_{j=0}^{n} \alpha_j k_{ij} u_j\right]\triangle t$$

$$u(x_i) - \lambda\left[\sum_{j=0}^{n} \alpha_j k_{ij} u_j\right]\triangle t = f(x_i)$$

$$u(x_i) - \lambda\Big[\alpha_0 k_{i0} u_0 + \alpha_1 k_{i1} u_1 + ... + \alpha_n k_{in} u_n\Big]\triangle t = f(x_i)$$

$$u(x_i) - \Big[\lambda\alpha_0 k_{i0} u_0 + \lambda\alpha_1 k_{i1} u_1 + ... + \lambda\alpha_n k_{in} u_n\Big]\triangle t = f(x_i)$$

for $i = 0$ to $i = n$

$$u(x_0) - \left[\lambda\alpha_0 k_{0,0} u_0 + \lambda\alpha_1 k_{0,1} u_1 + ... + \lambda\alpha_n k_{0,n} u_n\right] \triangle t = f(x_0)$$

$$u(x_1) - \left[\lambda\alpha_0 k_{1,0} u_0 + \lambda\alpha_1 k_{1,1} u_1 + ... + \lambda\alpha_n k_{1,n} u_n\right] \triangle t = f(x_1)$$

$$u(x_n) - \left[\lambda\alpha_0 k_{n,0} u_0 + \lambda\alpha_1 k_{n,1} u_1 + ... + \lambda\alpha_n k_{n,n} u_n\right] \triangle t = f(x_n)$$

$$\begin{pmatrix} u(x_0) \\ u(x_1) \\ . \\ . \\ . \\ u(x_n) \end{pmatrix} - \begin{pmatrix} \lambda\alpha_0 k_{00} u_0 + \lambda\alpha_1 k_{01} u_1 + .. + \lambda\alpha_n k_{0n} u_n \\ \lambda\alpha_0 k_{10} u_0 + \lambda\alpha_1 k_{11} u_1 + .. + \lambda\alpha_n k_{1n} u_n \\ . \\ . \\ . \\ \lambda\alpha_0 k_{n0} u_0 + \lambda\alpha_1 k_{n1} u_1 + .. + \lambda\alpha_n k_{nn} u_n \end{pmatrix} \triangle t = \begin{pmatrix} f(x_0) \\ f(x_1) \\ . \\ . \\ . \\ f(x_n) \end{pmatrix}$$

$$\begin{pmatrix} u_0 \\ u_1 \\ . \\ . \\ . \\ u_n \end{pmatrix} - \begin{pmatrix} \lambda\alpha_0 k_{00} & \lambda\alpha_1 k_{01} & .... & \lambda\alpha_n k_{0n} \\ \lambda\alpha_0 k_{10} & \lambda\alpha_1 k_{11} & .... & \lambda\alpha_n k_{1n} \\ & \cdots & & \\ & \cdots & & \\ & \cdots & & \\ \lambda\alpha_0 k_{n0} & \lambda\alpha_1 k_{n1} & .... & \lambda\alpha_n k_{nn} \end{pmatrix} \begin{pmatrix} u_0 \\ u_1 \\ . \\ . \\ . \\ u_n \end{pmatrix} \triangle t = \begin{pmatrix} f(x_0) \\ f(x_1) \\ . \\ . \\ . \\ f(x_n) \end{pmatrix}$$

$$\begin{pmatrix} u_0 \\ u_1 \\ . \\ . \\ . \\ u_n \end{pmatrix} - \begin{pmatrix} \lambda\alpha_0 k_{00}\triangle t & \lambda\alpha_1 k_{01}\triangle t & .... & \lambda\alpha_n k_{0n}\triangle t \\ \lambda\alpha_0 k_{10}\triangle t & \lambda\alpha_1 k_{11}\triangle t & .... & \lambda\alpha_n k_{1n}\triangle t \\ & . & & \\ & . & & \\ & . & & \\ \lambda\alpha_0 k_{n0}\triangle t & \lambda\alpha_1 k_{n1}\triangle t & .... & \lambda\alpha_n k_{nn}\triangle t \end{pmatrix} \begin{pmatrix} u_0 \\ u_1 \\ . \\ . \\ . \\ u_n \end{pmatrix} = \begin{pmatrix} f(x_0) \\ f(x_1) \\ . \\ . \\ . \\ f(x_n) \end{pmatrix}$$

$$\begin{pmatrix} 1-\lambda\alpha_0 k_{00}\triangle t & -\lambda\alpha_1 k_{01}\triangle t & .. & \lambda\alpha_n k_{0n}\triangle t \\ -\lambda\alpha_0 k_{10}\triangle t & 1-\lambda\alpha_1 k_{11}\triangle t & .. & -\lambda\alpha_n k_{1n}\triangle t \\ . & & & \\ . & & & \\ . & & & \\ -\lambda\alpha_0 k_{n0}\triangle t & -\lambda\alpha_1 k_{n1}\triangle t & .. & 1-\lambda\alpha_n k_{nn}\triangle t \end{pmatrix} \begin{pmatrix} u_0 \\ u_1 \\ . \\ . \\ . \\ u_n \end{pmatrix} = \begin{pmatrix} f_0 \\ f_1 \\ . \\ . \\ . \\ f_n \end{pmatrix}$$

$$\sum_{i=0}^{n}\sum_{j=0}^{n}(\delta_{ij}-\lambda\alpha_j k_{ij}\triangle t) = \begin{pmatrix} 1-\lambda\alpha_0 k_{00}\triangle t & .. & .. & \lambda\alpha_n k_{0n}\triangle t \\ -\lambda\alpha_0 k_{10}\triangle t & .. & .. & -\lambda\alpha_n k_{1n}\triangle t \\ . & & & \\ . & & & \\ . & & & \\ -\lambda\alpha_0 k_{n0}\triangle t & .. & .. & 1-\lambda\alpha_n k_{nn}\triangle t \end{pmatrix}$$

where $\delta_{ij} = \begin{cases} 1, i=j \\ 0, i\neq j \end{cases}$

Note

- When using the numerical quadrature approach, the solution will only exist when the determinant of the matrix is not zero.
- The α are called the weights. They are generated from the specific approximation formula being used.
- The $\triangle t$ are generated from the specific approximation formula being used.

Trapezoidal Rule

The trapezoidal rule for estimating definite integrals uses trapezoids to approximate the area under a curve.

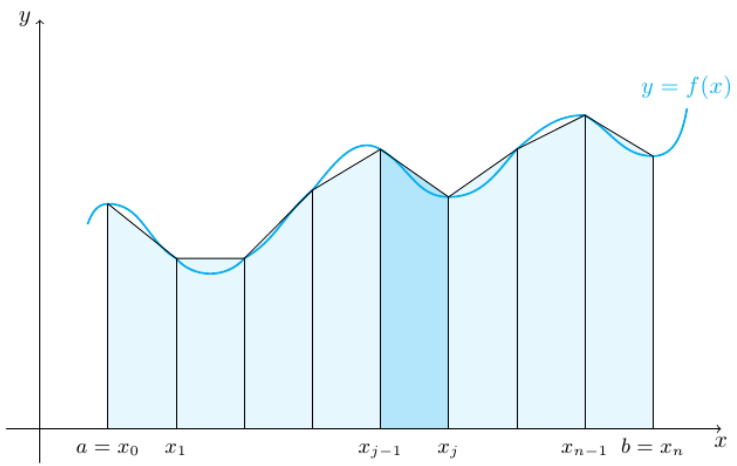

Figure: Trapezoidal Rule

$$\text{Area of a Trapezoid} = \frac{1}{2}h(b_1 + b_2), \tag{3}$$

where h = height of the trapezoid and b_1 and b_2 are the lengths of the parallel bases of the trapezoid.

Let the length of each sub-interval be given by $\triangle x$,

$\implies x_1 - x_0 = x_2 - x_1 = \cdots = x_j - x_{j-1} = \cdots = x_n - x_{n-1} = \triangle x$

From Figure 1, the height of the first trapezoid is $\triangle x$ and the parallel bases are of length $f(x_0)$ and $f(x_1)$. Thus, the area of the first trapezoid in Figure 1 is

$$\frac{1}{2}\triangle x\Big(f(x_0) + f(x_1)\Big) \tag{4}$$

Therefore, the areas of the remaining trapezoids are:

$$\frac{1}{2}\triangle x\Big(f(x_1) + f(x_2)\Big), \ldots, \frac{1}{2}\triangle x\Big(f(x_{j-1}) + f(x_j)\Big), \ldots,$$
$$\frac{1}{2}\triangle x\Big(f(x_{n-1}) + f(x_n)\Big) \tag{5}$$

Consequently,

$$\int_a^b f(x)dx \approx \frac{1}{2}\triangle x\Big(f(x_0)+f(x_1)\Big)+\frac{1}{2}\triangle x\Big(f(x_1)+f(x_2)\Big)+\cdots+$$
$$\frac{1}{2}\triangle x\Big(f(x_{j-1})+f(x_j)\Big)+\cdots+\frac{1}{2}\triangle x\Big(f(x_{n-1})+f(x_n)\Big) \tag{6}$$
$$\implies \int_a^b f(x)dx \approx \frac{1}{2}\triangle x\Big(f(x_0)+2f(x_1)+2f(x_2)+\cdots+$$
$$2f(x_{j-1})+2f(x_j)+\cdots+2f(x_{n-1})+f(x_n)\Big) \tag{7}$$

Trapezoidal Rule - General Formula

Assume that f(x) is continuous over [a,b]. Let n be a positive integer and $\triangle x = \dfrac{b-a}{n}$. Let [a,b] be divided into n sub-intervals, each of length $\triangle x$, with endpoints at $P = \{x_0, x_1, x_2, \ldots, x_n\}$. Let

$$T_n = \frac{\triangle x}{2}\left(f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right) \tag{8}$$

$$\therefore \lim_{n\to\infty} T_n = \int_a^b f(x)dx \tag{9}$$

Using the Trapezoidal rule for n=3

$$\int_{x_0}^{x_n} f(x) \approx \frac{h}{2}\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right]$$

$$\int_{x_0}^{x_n} f(x) \approx \frac{h}{2}\left[f(x_0) + 2\sum_{i=1}^{2} f(x_i) + f(x_3)\right]$$

$$\int_{x_0}^{x_n} f(x) \approx \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + f(x_3)\right]$$

Hence the corresponding α are

$$\alpha_{0,1,2,3} = 1, 2, 2, 1$$

and

$$\triangle t = \frac{h}{2}$$

Example

1. Use the trapezoidal rule with n=2 to estimate

$$\int_1^2 \frac{1}{x} dx$$

.

2. Use the trapezoidal rule to estimate the following integral using four sub-intervals:

$$\int_0^1 x^2 dx$$

.

Simpson's 1/3 Rule

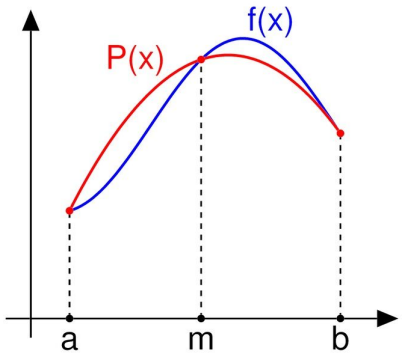

Figure: Simpson's 1/3 Rule

The equation of a curve passing through the points is given by:

$$y = ax^2 + bx + c \tag{10}$$

Therefore the area under a curve passing though points is given by:

$$\begin{aligned} \text{Area} &= \int_{-k}^{k} (ax^2 + bx + c)dx \\ &= \frac{ax^3}{3} + \frac{bx^2}{2} + cx \Big|_{-k}^{k} \end{aligned} \tag{11}$$

$$\begin{aligned}
\implies \text{Area} &= \frac{a(k)^3}{3} + \frac{b(k)^2}{2} + c(k) - \frac{a(-k)^3}{3} - \frac{b(-k)^2}{2} - c(-k) \\
&= \frac{ak^3}{3} + \frac{bk^2}{2} + ck + \frac{ak^3}{3} - \frac{bk^2}{2} + ck \\
&= \frac{2ak^3}{3} + 2ck \\
&= \frac{k}{3}(2ak^2 + 6c) \qquad (12)
\end{aligned}$$

From Figure 2, let $a = -k$ and $b = k$

$$\implies m = \frac{b+a}{2} = \frac{k+(-k)}{2} = 0$$

$\therefore y_0 = f(a), y_1 = f(m)$ and $y_2 = f(b)$

$$\implies y_0 = a(-k)^2 + b(-k) + c = ak^2 - bk + c \tag{13}$$

$$\implies y_1 = a(0)^2 + b(0) + c = c \tag{14}$$

$$\implies y_2 = a(k)^2 + b(k) + c = ak^2 + bk + c \tag{15}$$

$$\therefore y_2 + y_0 = ak^2 + bk + c + ak^2 - bk + c \tag{16}$$

$$= 2ak^2 + 2c$$

$$\text{but } c = y_1$$

$$\implies y_2 + y_0 = 2ak^2 + 2y_1 \tag{17}$$

$$\implies 2ak^2 = y_2 - 2y_1 + y_0 \tag{18}$$

$$\text{but Area} = \frac{k}{3}(2ak^2 + 6c)$$

$$\implies \text{Area} = \frac{k}{3}(y_2 - 2y_1 + y_0 + 6y_1)$$

$$= \frac{k}{3}(y_0 + 4y_1 + y_0) \tag{19}$$

Applying Equation (19) on a curve passing through more than three points; $X = \{x_0, x_1, x_2, \ldots, x_n\}$, where
$\triangle x = h = x_1 - x_0 = x_2 - x_1 = \cdots = x_j - x_{j-1} = \cdots = x_n - x_{n-1}$
$\implies Y = \{y_0, y_1, y_2, \ldots, y_n\} = \{f(x_0), f(x_1), f(x_2), \ldots, f(x_n)\}$

$$\therefore \text{Area} = \frac{h}{3}\Big(f(x_0) + 4f(x_1) + f(x_2)\Big) + \frac{h}{3}\Big(f(x_2) + 4f(x_3) + f(x_4)\Big)$$
$$+ \cdots + \frac{h}{3}\Big(f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big)$$
$$\implies \text{Area} = \frac{h}{3}\left(f(x_0) + \left[4\sum_{i=1,3,5,\ldots}^{n-1} f(x_i)\right] + \left[2\sum_{j=2,4,6,\ldots}^{n-2} f(x_j)\right] + f(x_n)\right) \tag{20}$$

- Using the Simpson's 1/3rd rule

$$\int_{x_0}^{x_n} f(x) \approx \frac{h}{3}\left[f(x_0) + 4\sum_{i=1,3,5,\ldots}^{n-2} f(x_i) + 2\sum_{i=2,4,\ldots}^{n-1} f(x_i) + f(x_n)\right]$$

$$\triangle t = \frac{h}{3}$$

Example

1. Use Simpson's 1/3rd rule with n = 4 to estimate

$$\int_1^2 e^{x^3} dx$$

.

2. Use the trapezoidal rule to approximate the following integral using eight sub-intervals:

$$\int_0^2 sin(\sqrt{x}) dx$$

.

Example

Solve the following integral equation using the numerical quadrature approach. Use Simpson's 3/8th rule and choose $n = 3$.

$$u(x) = -\int_0^1 xtu(t)\ dt + x^2$$

Solution:

$$\int_{x_0}^{x_n} f(x)dx = \frac{3h}{8}\left[f(x_0)+3\sum_{i=1,4,}^{n-2} f(x_i)+3\sum_{i=2,5,}^{n-1} f(x_i)+2\sum_{i=3,6,}^{n-3} f(x_i)+f(x_n)\right]$$

for $n = 3$

$$\int_{x_0}^{x_n} f(x)dx = \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right]$$

$$\lambda = -1, \qquad f(x) = x^2, \qquad k(x,t) = xt, \qquad \triangle t = \frac{3h}{8} = \frac{1}{8}$$

	0	1	2	3
x	0	1/3	2/3	1
t	0	1/3	2/3	1
f(x)	0	1/9	4/9	1
α	1	3	3	1

$$\begin{pmatrix} 1+\alpha_0 k_{00}\triangle t & \alpha_1 k_{01}\triangle t & \alpha_2 k_{02}\triangle t & \alpha_3 k_{03}\triangle t \\ \alpha_0 k_{10}\triangle t & 1+\alpha_1 k_{11}\triangle t & \alpha_2 k_{12}\triangle t & \alpha_3 k_{13}\triangle t \\ \alpha_0 k_{20}\triangle t & \alpha_1 k_{21}\triangle t & 1+\alpha_2 k_{22}\triangle t & \alpha_3 k_{23}\triangle t \\ \alpha_0 k_{30}\triangle t & \alpha_1 k_{31}\triangle t & \alpha_2 k_{32}\triangle t & 1+\alpha_3 k_{33}\triangle t \end{pmatrix} \begin{pmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \end{pmatrix} =$$

$$\begin{pmatrix} f_0 \\ f_1 \\ f_2 \\ f_3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 25/24 & 1/12 & 1/24 \\ 0 & 1/12 & 7/6 & 1/12 \\ 0 & 1/8 & 1/4 & 9/8 \end{pmatrix} \begin{pmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1/9 \\ 4/9 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 7/144 \\ 23/72 \\ 13/16 \end{pmatrix}$$

$$u(x) = f(x) + \lambda g(x) \int_a^b h(t)u(t)\ dt$$

$$u(x) = f(x) + \lambda g(x) \Big[\alpha_0 h_0 u_0 + \alpha_1 h_1 u_1 + \alpha_2 h_2 u_2 + \alpha_3 h_3 u_3 \Big] \triangle t$$

$$u(x) = x^2 - x \left[0 + \frac{7}{144} + \frac{23}{36} + \frac{13}{16} \right] \frac{1}{8} = x^2 - \frac{3x}{16}$$

ALTERNATIVELY

$$u(x) = -\int_0^1 xtu(t)\ dt + x^2$$

$$\int_{x_0}^{x_n} f(x)dx = \frac{3h}{8}\left[f(x_0)+3\sum_{i=1,4,}^{n-2} f(x_i)+3\sum_{i=2,5,}^{n-1} f(x_i)+2\sum_{i=3,6,}^{n-2} f(x_i)+f(x_n)\right]$$

for $n = 3$

$$\int_{x_0}^{x_n} f(x)dx = \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right]$$

$$\lambda = -1, \qquad f(x) = x^2, \qquad k(x,t) = xt, \qquad \triangle t = \frac{3h}{8} = \frac{1}{8}$$

	0	1	2	3
x	0	1/3	2/3	1
t	0	1/3	2/3	1
f(x)	0	1/9	4/9	1
α	1	3	3	1

$$u(x) = f(x) + \lambda g(x)[\sum_{j=0}^{n=3} \alpha_j h_j u_j]\triangle t$$

$$u(x) = f(x) - g(x)[\alpha_0 h_0 u_0 + \alpha_1 h_1 u_1 + \alpha_2 h_2 u_2 + \alpha_3 h_3 u_3]\triangle t$$

$$u(x) = f(x) - x[h_0 u_0 + 3h_1 u_1 + 3h_2 u_2 + h_3 u_3]\frac{1}{8}$$

Calculating for the various u

$$u(x_i) = f(x_i) - x_i[h_0 u_0 + 3h_1 u_1 + 3h_2 u_2 + h_3 u_3]\frac{1}{8}$$

$$u_i = f(x_i) - x_i[h_0u_0 + 3h_1u_1 + 3h_2u_2 + h_3u_3]\frac{1}{8}$$

$$u_0 = f(x_0) - x_0[h_0u_0 + 3h_1u_1 + 3h_2u_2 + h_3u_3]\frac{1}{8}$$

$$u_0 = 0 - 0\left[0u_0 + 3 \times \frac{1}{3}u_1 + 3 \times \frac{2}{3}u_2 + u_3\right]\frac{1}{8} = 0 \tag{21}$$

$$u_1 = f(x_1) - x_1[h_0u_0 + 3h_1u_1 + 3h_2u_2 + h_3u_3]\frac{1}{8}$$

$$u_1 = \frac{1}{9} - \frac{1}{3}\left[0u_0 + 3 \times \frac{1}{3}u_1 + 3 \times \frac{2}{3}u_2 + u_3\right]\frac{1}{8}$$

$$\frac{25}{24}u_1 + \frac{1}{12}u_2 + \frac{1}{24}u_3 = \frac{1}{9} \tag{22}$$

$$u_2 = f(x_2) - x_2[h_0u_0 + 3h_1u_1 + 3h_2u_2 + h_3u_3]\frac{1}{8}$$

$$u_2 = \frac{4}{9} - \frac{2}{3}\left[0u_0 + 3 \times \frac{1}{3}u_1 + 3 \times \frac{2}{3}u_2 + u_3\right]\frac{1}{8}$$

$$\frac{1}{3}u_1 + \frac{7}{3}u_2 + \frac{1}{3}u_3 = \frac{4}{\ } \tag{23}$$

$$u_3 = f(x_3) - x_3[h_0 u_0 + 3h_1 u_1 + 3h_2 u_2 + h_3 u_3]\frac{1}{8}$$

$$u_3 = 1 - \left[0u_0 + 3 \times \frac{1}{3}u_1 + 3 \times \frac{2}{3}u_2 + u_3\right]\frac{1}{8}$$

$$\frac{1}{8}u_1 + \frac{1}{4}u_2 + \frac{9}{8}u_3 = 1 \tag{24}$$

Combining (4),(5),(6)

$$\begin{pmatrix} \frac{25}{24} & \frac{1}{12} & \frac{1}{24} \\ \frac{1}{12} & \frac{7}{6} & \frac{1}{12} \\ \frac{1}{8} & \frac{1}{4} & \frac{9}{8} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{9} \\ \frac{4}{9} \\ 1 \end{pmatrix}$$

Using Cramer's rule

$$u_1 = \begin{vmatrix} \frac{1}{9} & \frac{1}{12} & \frac{1}{24} \\ \frac{4}{9} & \frac{7}{6} & \frac{1}{12} \\ 1 & \frac{1}{4} & \frac{9}{8} \end{vmatrix} \div \begin{vmatrix} \frac{25}{24} & \frac{1}{12} & \frac{1}{24} \\ \frac{1}{12} & \frac{7}{6} & \frac{1}{12} \\ \frac{1}{8} & \frac{1}{4} & \frac{9}{8} \end{vmatrix} = \frac{7}{144}$$

$$u_2 = \begin{vmatrix} \frac{25}{24} & \frac{1}{9} & \frac{1}{24} \\ \frac{1}{12} & \frac{4}{9} & \frac{1}{12} \\ \frac{1}{8} & 1 & \frac{9}{8} \end{vmatrix} \div \begin{vmatrix} \frac{25}{24} & \frac{1}{12} & \frac{1}{24} \\ \frac{1}{12} & \frac{7}{6} & \frac{1}{12} \\ \frac{1}{8} & \frac{1}{4} & \frac{9}{8} \end{vmatrix} = \frac{23}{72}$$

$$u_3 = \begin{vmatrix} \frac{25}{24} & \frac{1}{12} & \frac{1}{9} \\ \frac{1}{12} & \frac{7}{6} & \frac{4}{9} \\ \frac{1}{8} & \frac{1}{4} & 1 \end{vmatrix} \div \begin{vmatrix} \frac{25}{24} & \frac{1}{12} & \frac{1}{24} \\ \frac{1}{12} & \frac{7}{6} & \frac{1}{12} \\ \frac{1}{8} & \frac{1}{4} & \frac{9}{8} \end{vmatrix} = \frac{13}{16}$$

$$u(x) = f(x) - x[h_0 u_0 + 3h_1 u_1 + 3h_2 u_2 + h_3 u_3]\frac{1}{8}$$

but $h_0 = 0, h_1 = \frac{1}{3}, h_2 = \frac{2}{3}$ and $h_3 = 1$

$$\implies u(x) = x^2 - \frac{x}{8}[u_1 + 2u_2 + u_3]$$

$$u(x) = x^2 - \frac{x}{8}\left[\frac{7}{144} + 2 \times \frac{23}{72} + \frac{13}{16}\right]$$

$$u(x) = x^2 - \frac{x}{8}\left[\frac{3}{2}\right] = x^2 - \frac{3x}{16}$$

Exercises

Solve the following integral equation using the numerical quadrature approach. Use Simpson's 1/3rd rule and the Trapezoidal rule. Choose a suitable value for **n**

$$u(x) = -\int_0^1 xtu(t)\ dt + x^2$$